HANSANA PABASHWARA

COMPUTER SCIENCE AND ENGINEERING UNDERGRADUATE

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SUMMARY

Software Engineer with a strong background in Computer Science and full-stack development. Experienced in building scalable web and mobile applications, from FinTech systems to AI-powered project management tools. Dedicated to solving complex architectural challenges through clean code and innovative, data-driven solutions.

EDUCATION

BSc Eng Hons, Computer Science and Engineering **2022 - 2026**
University of Moratuwa

- Stream: Data Science and Engineering | GPA: 3.74
- Honors: Winner of the Brainstorm Healthcare Innovation Competition

PROFESSIONAL EXPERIENCE

Software Engineer Intern | IronOne Technologies **Dec 2024 – June 2025**

- End-to-End System Development: Developed a Primary Dealer system for Treasury Bills and Bonds, managing the full lifecycle of backend and frontend operations.
- Architecture & Security: Designed and implemented secure authentication management and robust database operations using Keycloak and PostgreSQL.
- Full-Stack Implementation: Built responsive interfaces and high-performance APIs using React, TypeScript, and Quarkus.

PROJECTS

AgriGrid | Farmer Identification System

- Developing a microservices-based farmer identification platform using Spring Boot, designed to support large-scale data processing and system integration.
- Implementing centralized authentication and authorization using Keycloak, with a React frontend and PostgreSQL database.
- Containerizing services using Docker to ensure consistent development, deployment, and production environments.

ORIGO DB | Database Engine Development

- Developing a relational database engine from scratch in C, focusing on low-level memory management and file I/O.Hybrid Storage Strategy: Implementing a Redis-based Risk Engine for sub-millisecond balance validation and a PostgreSQL ledger for financial data persistence.
- Engineering custom data structures and storage engines to handle raw data persistence and retrieval efficiency.

ProjecX | Agentic AI Project Management Tool |

- Architected an automated task management system using FastAPI, Next.js, and AI agents.
- Managed cloud infrastructure on Azure and Vercel with PostgreSQL for data persistence.

ArcFuse | Remote Sensing Data Fusion System

- Architecting a multi-modal data fusion pipeline using Vision Transformers (ViT) and U-Net for large-scale feature detection in the Pan-Arctic region.
- Optimizing high-volume data ingestion of satellite imagery, location embeddings, and digital elevation models using PyTorch and OpenCV and Rasterio.
- Managing complex geospatial datasets and spatial queries using QGIS and custom data processing scripts.

EduGenius | RAG-based Mathematics Tutoring Web App

- Architected a web-based Retrieval-Augmented Generation (RAG) chatbot using LangChain, Gemini, and ChromaDB.Optimizing high-volume data ingestion of satellite imagery, location embeddings, and digital elevation models using PyTorch and OpenCV and Rasterio.
- Engineered a specialized mathematics question and answer dataset by implementing OCR-based data extraction to improve retrieval accuracy.
- Developed the application interface using Streamlit and managed data persistence via MongoDB.

Full-Stack Blog Application | Next.js & tRPC

- Architected a modern blog platform using Next.js and tRPC to achieve end-to-end type safety between the client and serverEngineered a specialized mathematics question and answer dataset by implementing OCR-based data extraction to improve retrieval accuracy.
- Developed a highly responsive UI with Tailwind CSS and managed database interactions using Prisma ORM.
- Ensured robust data handling and seamless API integration by leveraging tRPC's procedure-based communication.

Quizzy | Cross-Platform Mobile Application

- Developed a mobile quiz application using Flutter and Dart, featuring real-time data synchronization with Firebase Firestore.
- Integrated Firebase Authentication and specialized Data Transfer Objects (DTOs) to ensure secure and structured data flow.
- Designed a responsive UI/UX with state-managed components for an enhanced user experience.

TECHNICAL SKILLS

- **Languages**
  - C, Java, Python, SQL, C++, Dart, JavaScript, TypeScript
- **Backend & Systems**
  - Spring Boot, Quarkus, Node.js, FastAPI, tRPC, Redis, Docker
- **Frontend & Mobile**
  - React, Next.js, HTML/CSS, Tailwind, Flutter, Firebase
- **Databases**
  - PostgreSQL, MySQL, SQLite, MongoDB, ChromaDB, Prisma, HBase, DynamoDB

ADDITIONAL INFORMATION

- **Languages**
  - English
  - Sinhala
- **Certifications**
  - NVIDIA Fundamentals of Deep Learning.
  - AWS Academy Data Engineering.
  - AWS Academy Machine Learning Foundations.
  - Google Cloud Machine Learning Operations (MLOps)
  - LangChain for LLM Application Development By DeepLearning.AI